

Elena Gusarevich

Python Backend Developer / Python Automation Engineer

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PROFESSIONAL SUMMARY

Python Backend Developer / Python Automation Engineer focused on backend automation, security workflows, API integrations, scraping infrastructure, and production Python systems.

Experienced in building modular backend services, async processing pipelines, threat intelligence enrichment workflows, anti-bot automation, and resilient analytics systems.

Recently delivered Microsoft Sentinel automation projects involving MSTICPy, KQL orchestration, Azure ML notebooks, cross-domain correlation, Isolation Forest anomaly detection, SHAP explainability, and incident automation workflows.

Strong in API debugging, HTTP session analysis, clean architecture, maintainable business logic, and production-ready automation solutions.

CORE EXPERTISE

Backend Development, Python Automation, Security Automation, API Debugging, HTTP Session Analysis, Distributed Systems, ETL Pipelines, Threat Intelligence Enrichment, Incident Automation, Async Processing, Background Workers, Integration Testing, Structured Analytics, Scraping Infrastructure, and Production Infrastructure.

TECHNICAL SKILLS

Experienced in building backend automation systems, analytics pipelines, data-processing workflows, asynchronous services, and production-ready distributed systems for integration-heavy applications.

Core: Python 3.10+, AsyncIO, OOP, Clean Architecture

Backend: FastAPI, SQLAlchemy, Alembic, REST APIs, async workers, service layer architecture, repository pattern, background jobs

HTTP & APIs: requests, httpx, REST APIs, sessions, cookies, headers

Automation: SeleniumBase (UC Mode), Playwright, browser automation, scraping infrastructure, HTTP session debugging

Data & Messaging: PostgreSQL, MongoDB, Kafka, ETL pipelines, data processing, asynchronous processing

Cloud & Infra: Docker, Kubernetes, Helm, Linux, Systemd, MinIO/S3, Azure ML

Languages: Python, SQL, JavaScript, Kotlin

Mobile: Android, Gradle

Security & TI: MSTICPy, WHOIS, DNSBL, GeolIP, URLhaus, MalwareBazaar

Analytics: pandas, Plotly, SHAP, Isolation Forest, merge_asof

Analytics & Automation: anomaly detection, structured analytics, risk scoring, decision workflows

Testing: pytest, pytest-asyncio, pytest-cov, integration testing

WORK EXPERIENCE

Python Backend Developer

Jun 2026 – Present

QLAN

Python Backend Developer working on a production media processing and social-platform data extraction system.

- **Backend Development:** Develop and maintain backend services for automated media processing, metadata extraction, and statistics collection across social media platforms.
- **Data Processing:** Build asynchronous backend services and data-processing pipelines using FastAPI, SQLAlchemy, PostgreSQL, Kafka, and Alembic.
- **Infrastructure:** Work with Docker, Kubernetes, Helm, Linux, MinIO/S3, and distributed worker infrastructure.
- **Production Reliability:** Focus on production reliability, fault tolerance, graceful recovery, and maintainability.

Python Automation Engineer | Backend Developer

Jan 2024 – Jun 2026

Private Practice

- **Backend Automation:** Built automation systems for data extraction, API integrations, background processing, and structured reporting.
- **Scraping & Anti-Bot Systems:** Developed scraping and browser automation workflows for dynamic websites, protected flows, and API-based data extraction.
- **HTTP/API Debugging:** Investigated complex request issues involving sessions, cookies, headers, payload structure, redirects, and browser-vs-script differences.

- **Infrastructure:** Deployed and maintained Linux VPS environments using Docker, Systemd services, background workers, and MongoDB/PostgreSQL storage.
- **Architecture:** Designed and developed modular Python systems with repository patterns, service layers, async workers, idempotent processing, and automated test coverage.

Email Analysis Engine for Microsoft Sentinel Python, Threat Intelligence, APIs, pytest 2026

- Delivered a modular phishing analysis engine for Microsoft Sentinel incident automation.
- Integrated WHOIS, DNSBL, GeolIP, reverse DNS, URLhaus, MalwareBazaar, homoglyph detection, and structured scoring.
- Built contract-stable JSON input/output, graceful timeout handling, external JSON scoring config, and production-ready reporting.
- Validated the engine in a real security automation workflow with unit and integration test coverage above the required threshold.

Threat Hunting Notebooks for Microsoft Sentinel & XDR Python, MSTICPy, pandas, Azure ML, SHAP 2026

- Implemented Python analytics layers for Microsoft Sentinel and XDR threat hunting notebooks in Azure ML environments.
- Built KQL orchestration utilities, Sentinel incident automation, MISP integrations, VirusTotal caching, and threat enrichment workflows.
- Developed cross-domain attack correlation pipelines across email, endpoint, Safe Links, and cloud activity using pandas merge_asof().
- Implemented AI-assisted anomaly detection workflows using Isolation Forest, SHAP explainability, and analyst-oriented risk scoring systems.
- Designed resilient notebook flows with graceful handling of sparse telemetry, unavailable services, and partial-stage failures.

Art Lead 3D | Senior Technical Artist Jan 2022 – Oct 2024 NDA Online PC Game / Game Studios

- **Automation Scripts:** Developed Python tools for Blender and Unreal Engine to batch-process assets and reduce manual production work.
- **Leadership:** Managed a cross-functional team of 15 specialists, including hiring, task estimation, documentation, and delivery control.
- **Technical Pipeline:** Improved production workflows between artists, designers, and technical teams.

* Prior to 2022: Successful career in Design Engineering & Art Management. Full history available on LinkedIn.

KEY PROJECTS

Ritma - Cross-Platform Notification Platform Python, AsyncIO, MongoDB, Docker, Kotlin, Android 2026

- Rebuilt an MVP into a modular backend platform with repository pattern, service layer, isolated adapters, and background workers.
- Implemented timezone-safe event modeling, notification orchestration, partner-aware flows, custom messaging logic, recipient preferences, and phase-based delivery rules.
- Designed an adapter-based architecture supporting multiple client and messaging interfaces, with idempotent processing across asynchronous workflows.
- Built an Android adapter in Kotlin and released the application on Google Play, extending the backend platform to a production mobile client.
- Maintained a comprehensive automated test suite with 784 tests and 91% test coverage.

SMS Verification & Anti-Bot Bypass System SeleniumBase, Manifest V3, API, Proxy Auth 2025

- Engineered a browser automation solution for protected registration flows using SeleniumBase UC Mode and custom Chrome extensions.
- Handled proxy authentication, CAPTCHA-solving services, SMS verification APIs, and React SPA input events.
- Improved automation reliability by emulating realistic browser behavior and debugging request/session issues.

Microsoft Sentinel Threat Hunting Notebooks Python, MSTICPy, Azure ML, pandas 2026

- Built Python analytics components for Microsoft Sentinel XDR threat hunting workflows in Azure ML.
- Implemented cross-domain correlation, KQL utilities, threat intelligence enrichment, and anomaly-detection support tooling.

EDUCATION

Russian Academy of Crafts 2013 – 2014 Artistic Forging (Third-class Artisan)

Additional self-directed specialization in Python backend development, automation engineering, scraping systems, API integrations, Linux infrastructure, and production automation.